



Computer Networks

计算机网络 (CST2398)

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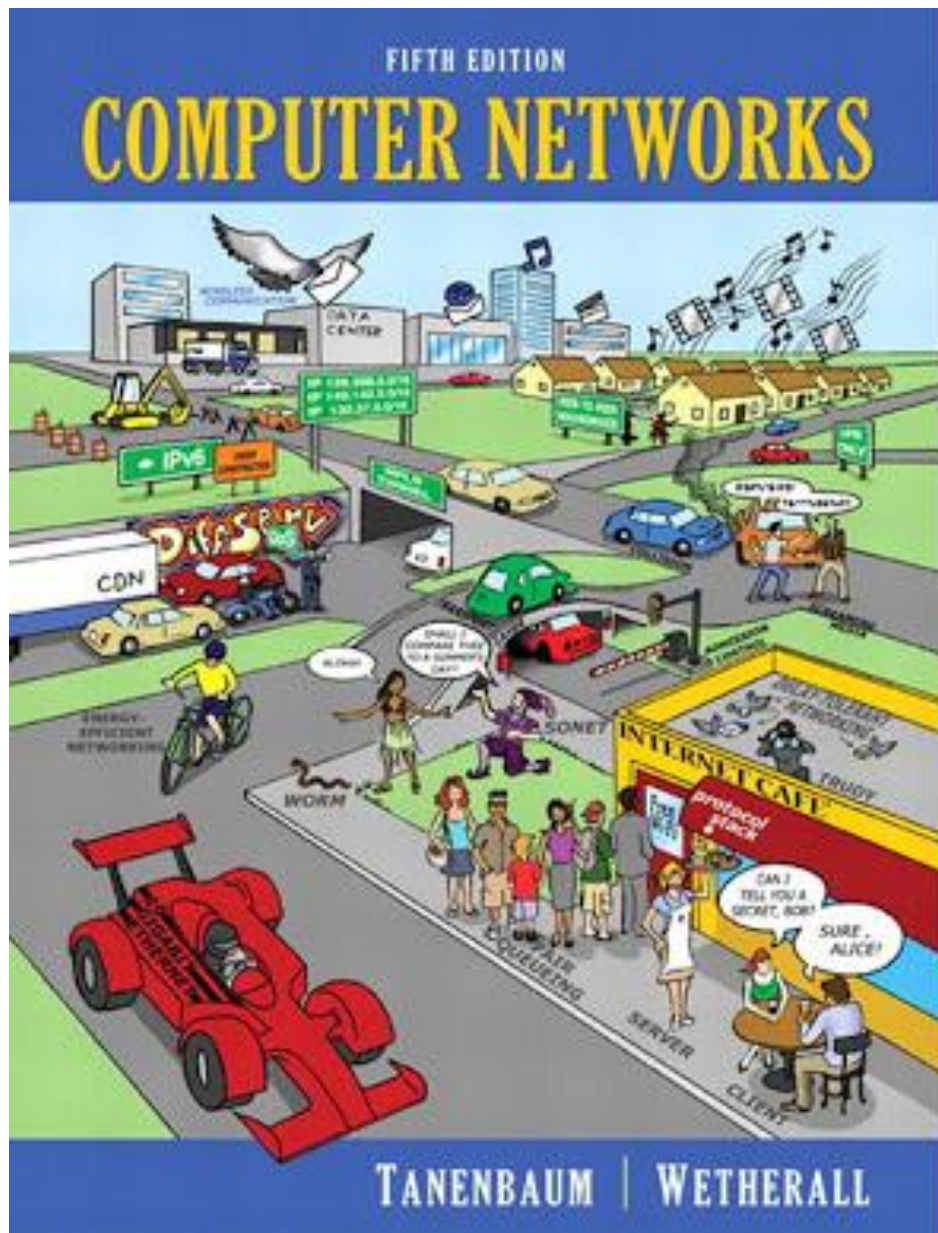


01

Introduction

Contents of this Chapter

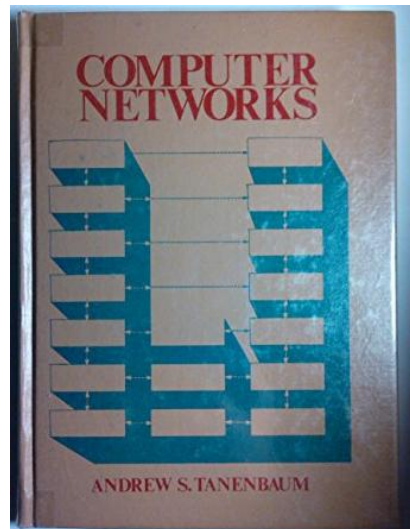
- *What is a **computer network**?*
- *Why do we need a '**network**'?*
- *Network Hardware: **classification** of computer networks*
- *Network Core: **circuit switching** v.s. **packet switching***
- *Network Software: **layered model** and **protocols***
- *History and the Future*
- *Standards and Organizations*



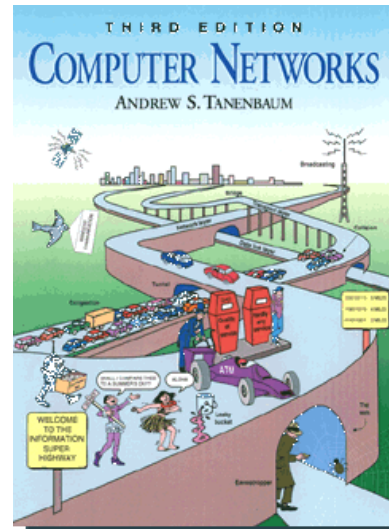
Computer Networks

- *Are computer systems.*
- **Def.** A collection of *autonomous* computers *interconnected* by a single technology.

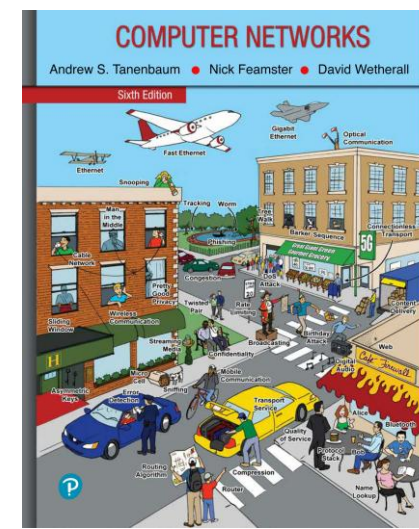
Prentice Hall, 1985



Prentice Hall, 1996



Pearson, 2021



A simple test by the end of this semester: Know all the concepts in the figure?

Why is a 'network' necessary?

- **Cheaper, smaller, and *much more* computers.**

*Specialized instruments
in a glass box*



*'Nine of the computers were
sold, mainly to national labs
and scientific users...'*

*Resource sharing
Files, data, processor, printer...*



**The network is
becoming a
computer.**



*Universal tool
everywhere*



*Information sharing
IM, video, games, cloud...*

Key Enablers of a Computer Network

- **Digitization** of Signals

Text/audio/video/image

Aa



Sample
Quantize
Encode

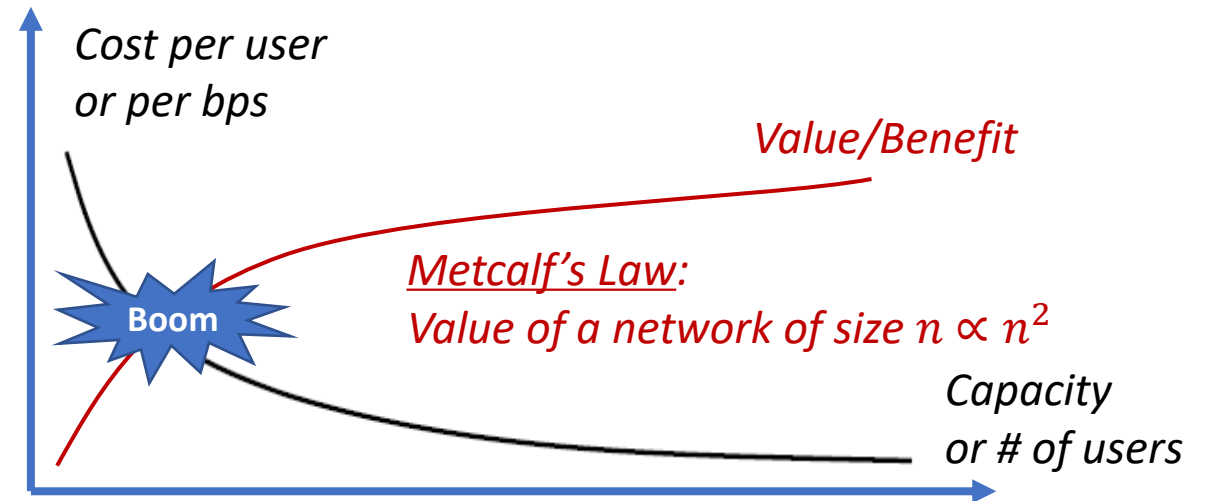
Binary stream



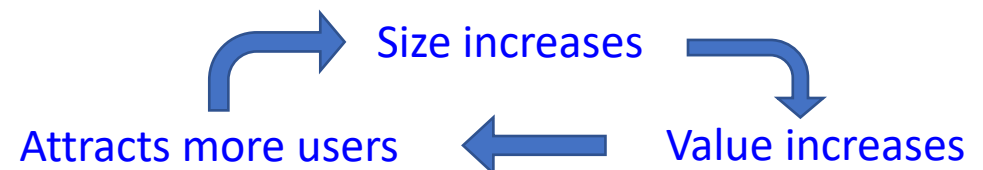
Storable, and suitable for
computer communication

- **Economies of Scale**

A network has (almost) fixed costs.



- **Network Externalities**



Classification of Computer Networks

- **By transmission technology:**

- Broadcasting/multi-casting, e.g. wireless
- point-to-point



Key diff.:
Medium



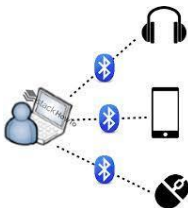
- **By scale/range:**

- The Internet (from ARPANET) – planet wide ‘the one, the only’
- **Wide Area Network (WAN)** – country to continent wide
cable, co-ax, fiber, satellite
- Metropolitan Area Network (MAN) – city wide
- **Local Area Network (LAN)** – ranges from several Km to 10 m



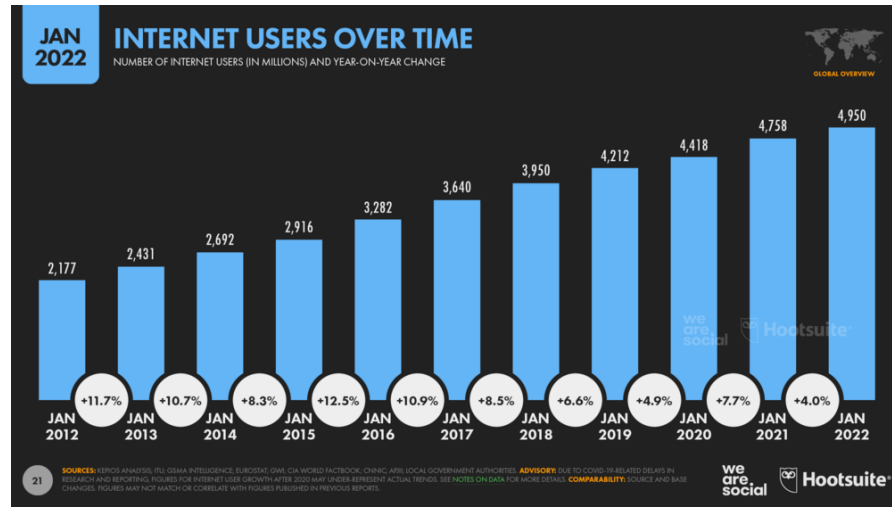
Ethernet/RJ45 cable, WiFi

- **Personal Area Network (PAN)** – <10 m
Bluetooth, Zigbee



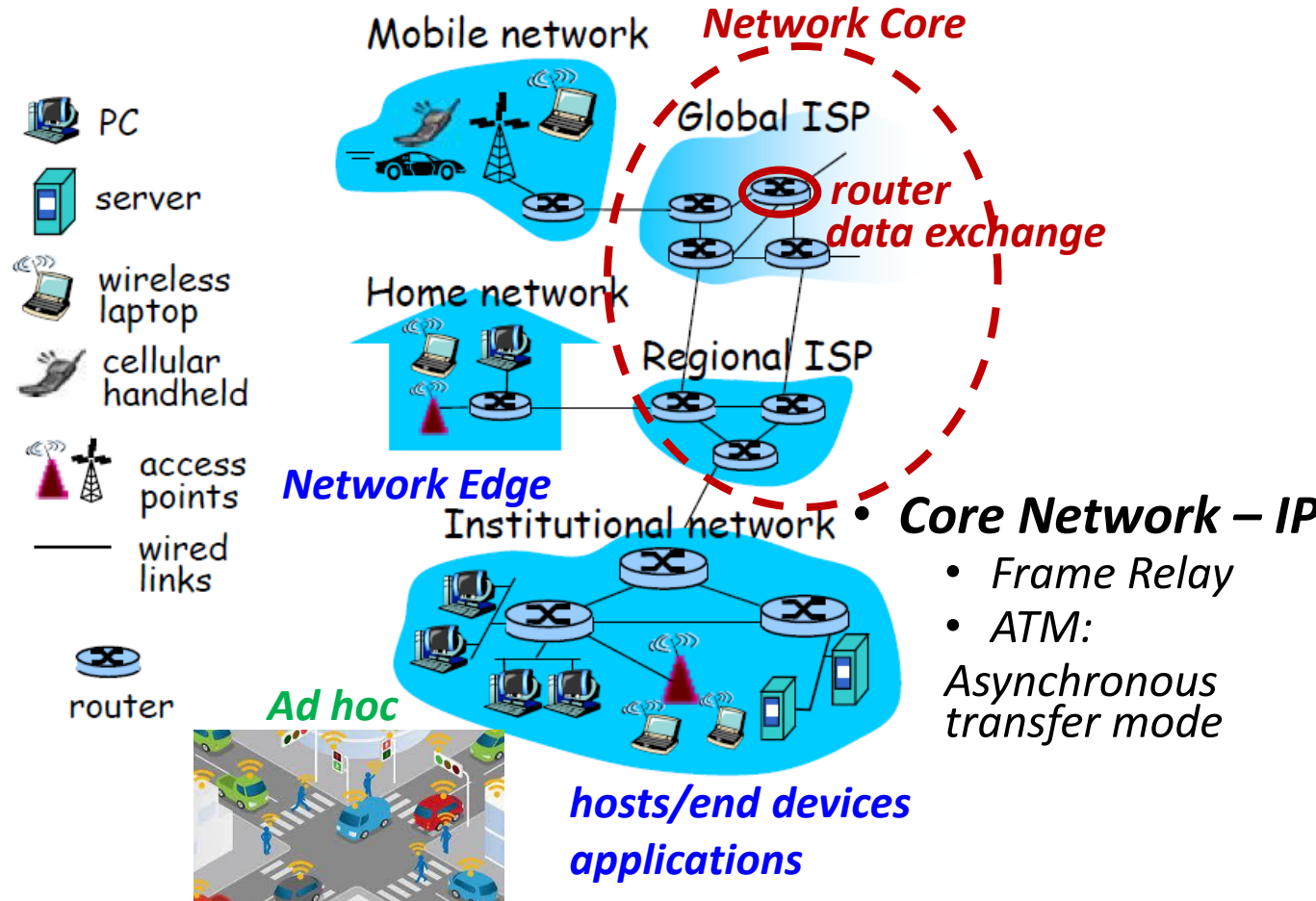
The Internet

- **Internet v.s. internet (network of networks)**

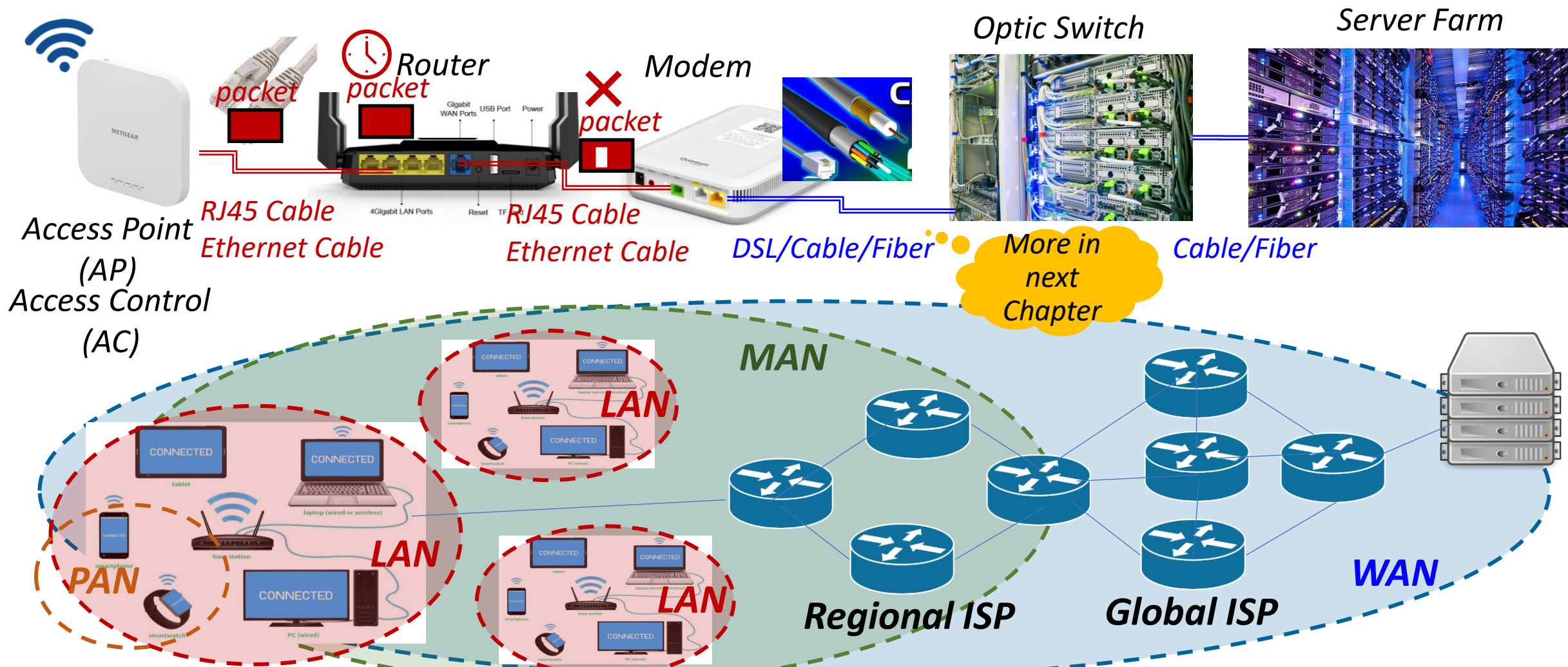


- **Access Network – Edge**

- ADSL: asymmetric digital subscriber line (high ↓, low ↑)
- FTTH: fiber-to-the-home
- Wireless: WiFi, cellular ...



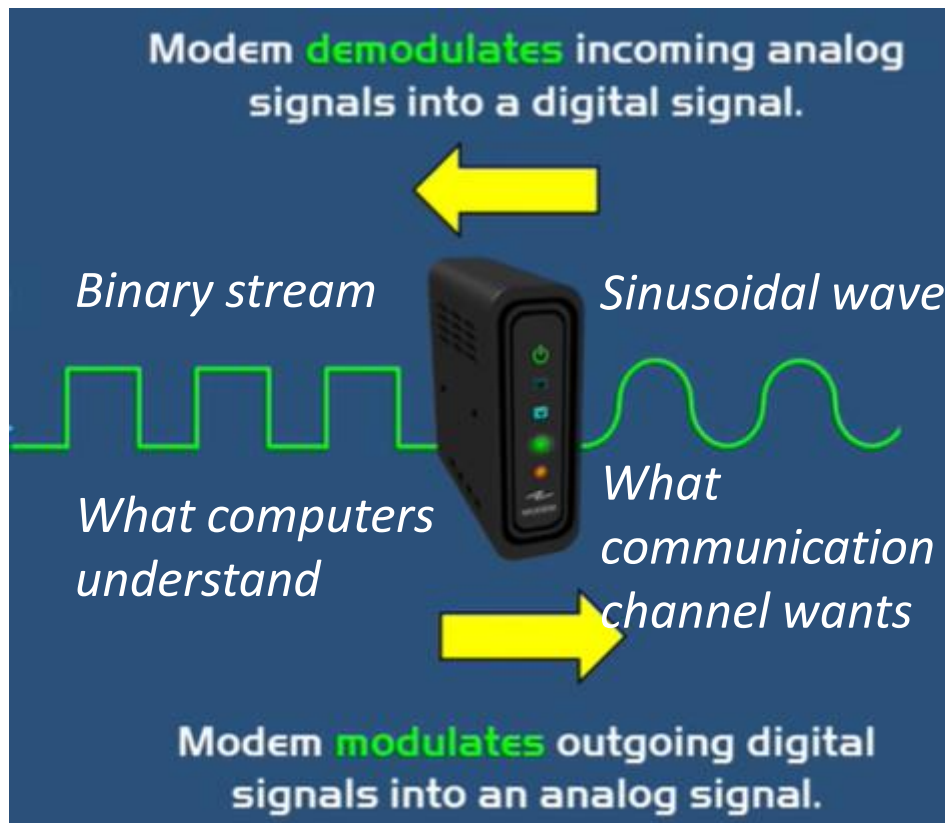
Connecting to the Internet



Hardware: Modem v.s. Router

We'll see more
in the next
chapter.

- **Modem** converts (digital from/to analog) signals.



- **Routers** connects (sub)networks, and exchanges packets.

- Don't need a router if there is only one device to connect.
- Don't need a router if there is only one network in the world. (Then what we need is a **switch/hub**).
- Sometimes router+modem are combined in one device.

Ex.1.1 Data Rate

- **Consider the old-schooled telephony network.**

- Audible voice ranges from 30 – 3400 Hz (roughly 0-4 KHz).

sample

- The voice signal is sampled 8000 times per second.

quantize

- Each voice sample is represented by 8 bit data.

We'll see why
in the next
chapter.



- **Find the data rate of each telephone line.**

- What is the **Baud rate**, or symbol rate?

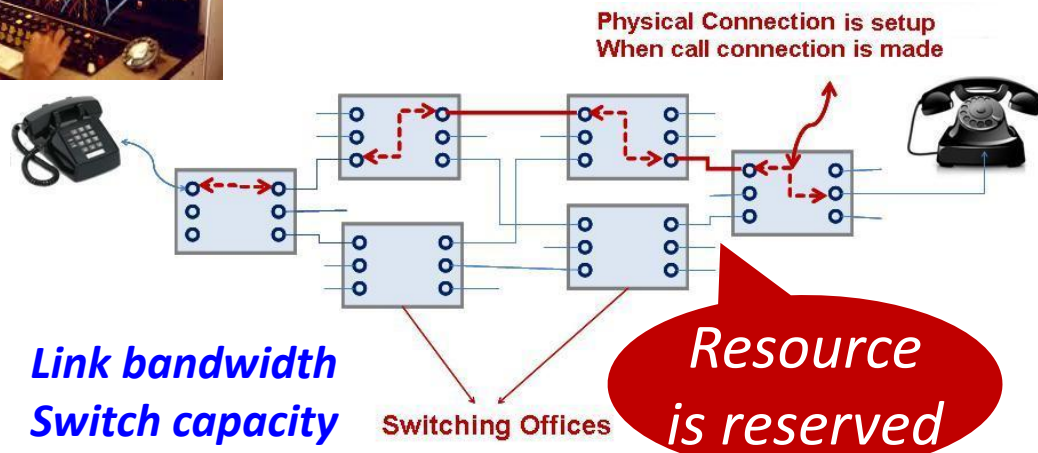
- What is the **bit rate**, or data rate?

Network Core: Switching

- **Core: interconnected routers**
- **How is data transferred through the network?**

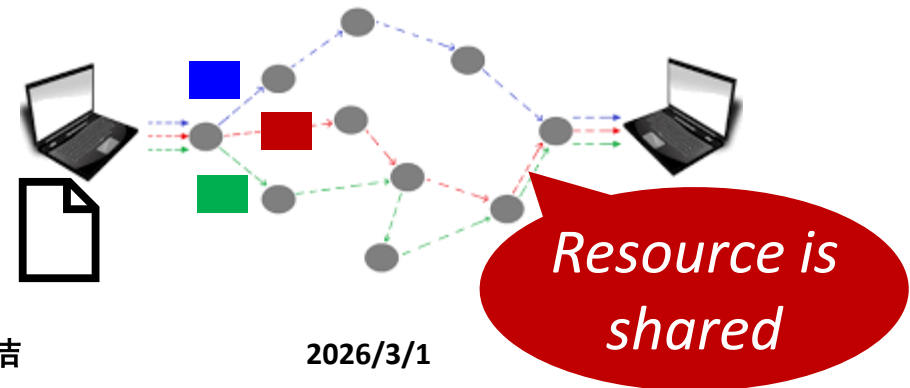
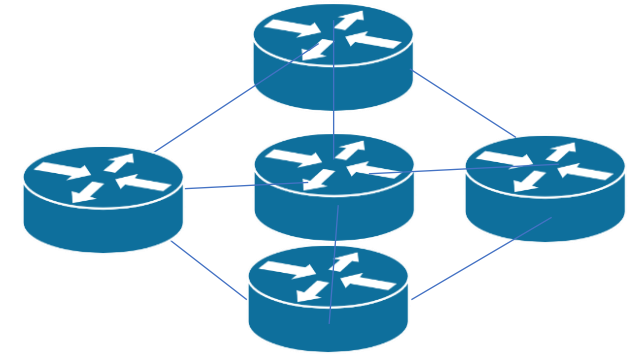
- **Circuit Switching**

- Construct a *dedicated communication path* between the two end devices



- **Packet Switching**

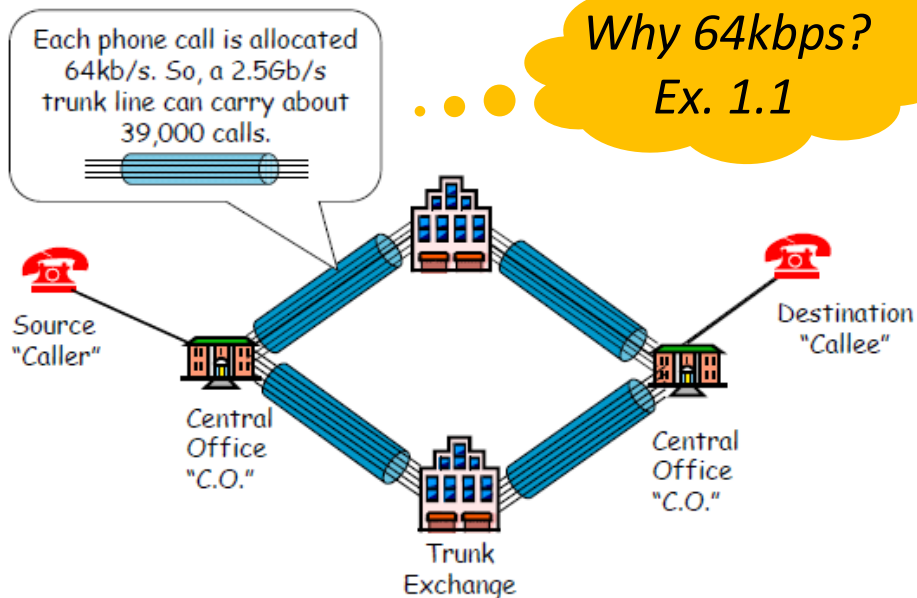
- Broke a message into a series of *packets*
- Each node (switch/router) decides how/where to transfer a packet



Circuit Switching

- **Steps**

- 1) Circuit establishes
- 2) Information transfer
 - Can be analog or digital
- 3) Circuit terminates



- **Advantages**

Once established,

- Network is transparent to users.
- Data **rate is fixed**/guaranteed.
- Data arrives **in order**.

- **Disadvantages**

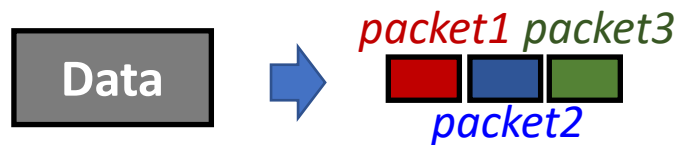
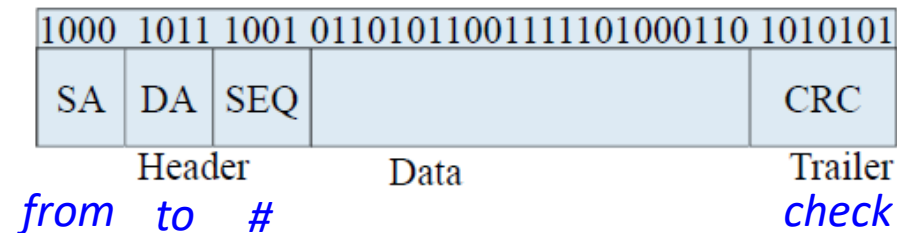
- Circuit establishment delay
- **Low utilization**
- Resource is divided into channels, and a channel is occupied throughout the connection.

How to divide?
Multiplexing

Packet Switching (1/2)

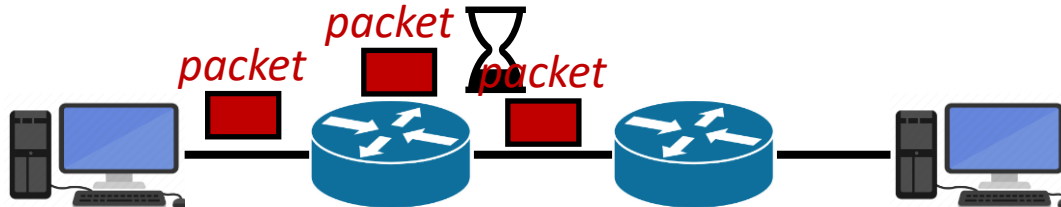
- **Steps**

- Data is divided into *packets*.



- **Store-and-forward**

A *packet* must be received as a whole (every bit) before it can be transmitted.



- **Advantages**

- Each packet uses the full link bandwidth.
- Resource is shared among users.
- Resource can be used *on demand*.
- Can admit *more users!!*

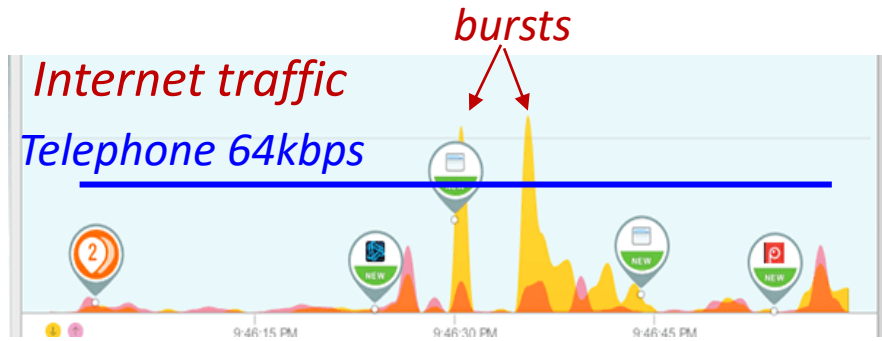
- **Disadvantages**

- Resource contention
- Congestion
- Routing is complex
- Packets may arrive out of order.

Why? We will show it soon

Packet Switching (2/2)

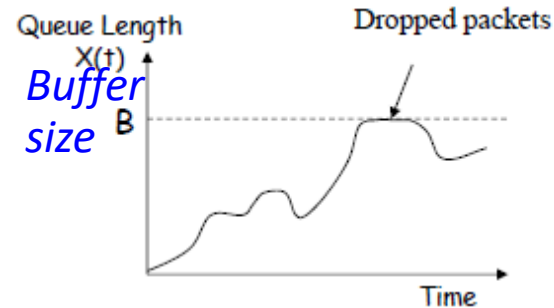
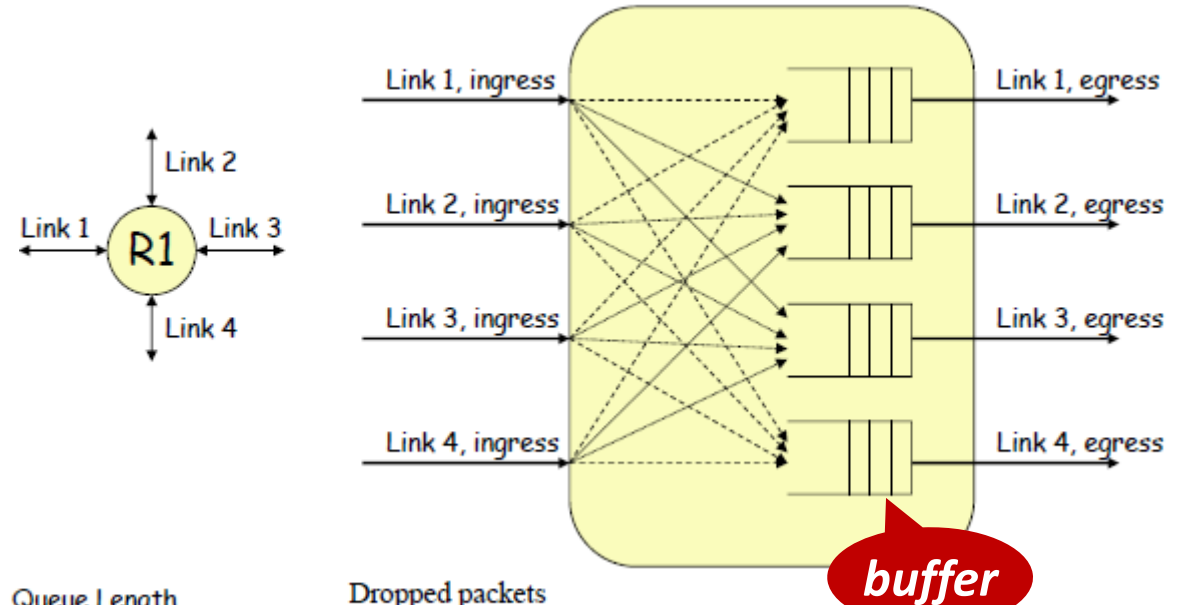
- **Bursty Data**



Packet Switching can handle **bursty data**:

- **Resource is shared.**
- **Resource is used on demand.**
- **Each packet uses the full bandwidth.**

- **Simple router model**

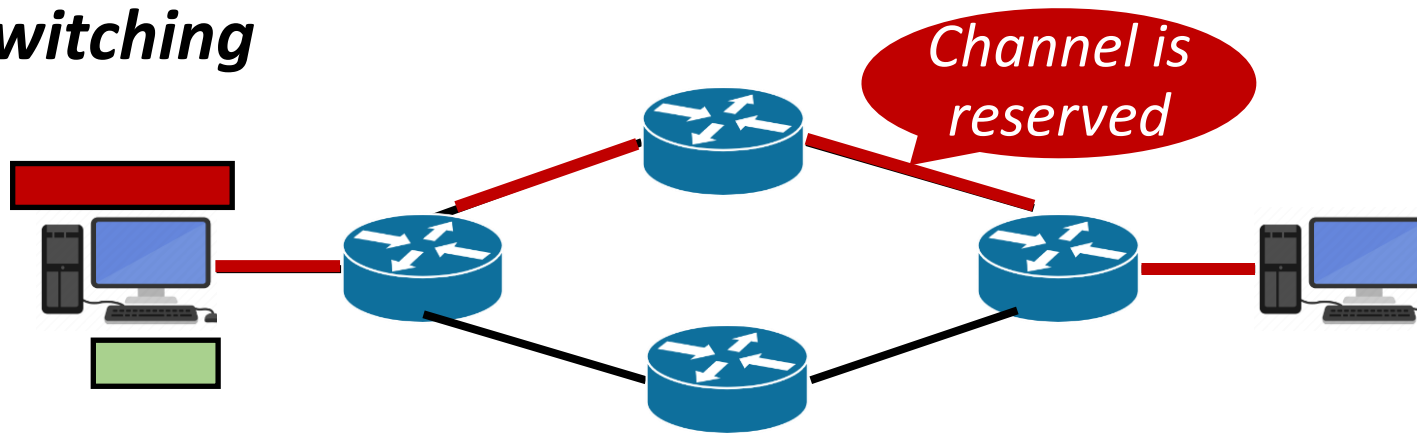


Buffer can absorb temporary **bursts**.

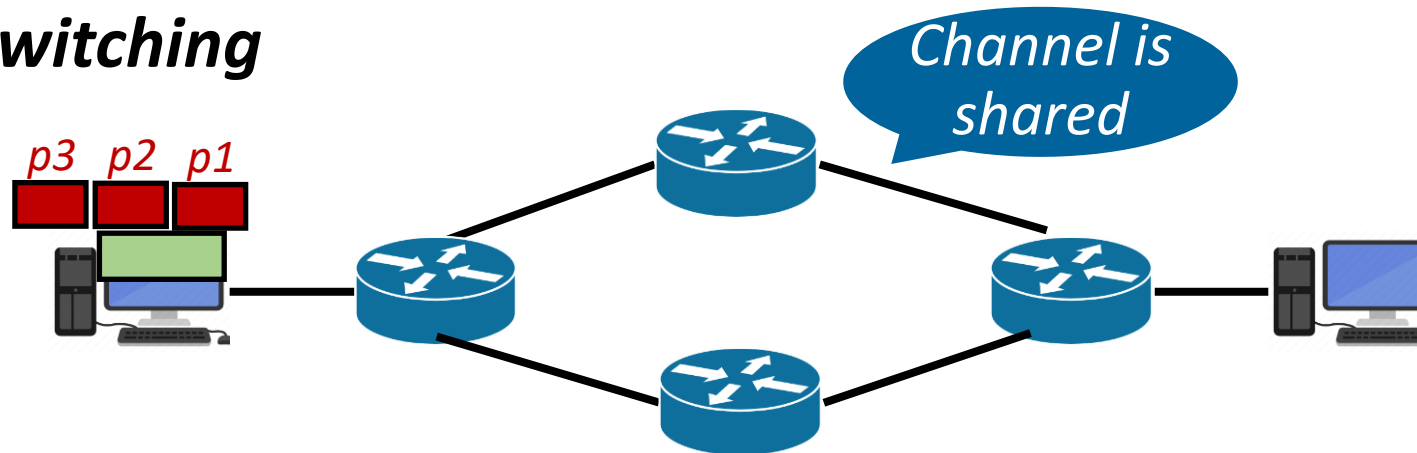
Queuing Theory

E.g. Circuit Switching v.s. Packet Switching

- **Circuit Switching**



- **Packet Switching**



Comparison of the Two

	<i>Circuit Switching</i>	<i>Packet Switching</i>
<i>Delay</i>	<i>Constant</i>	<i>Variable</i>
<i>Order</i>	<i>Data arrive in order</i>	<i>Packets may arrive out of order</i>
<i>BW Efficiency</i>	<i>Inefficient (dedicated)</i>	<i>Efficient (on-demand)</i>
<i>Routing</i>	<i>Simple</i>	<i>Complex</i>
<i>QoS</i>	<i>'All or nothing'</i>	<i>'Graceful degradation'</i>
<i>Control</i>	<i>Low complexity</i>	<i>High complexity</i>
<i>Scenario</i>	<i>Voice communication</i>	<i>Data communication</i>

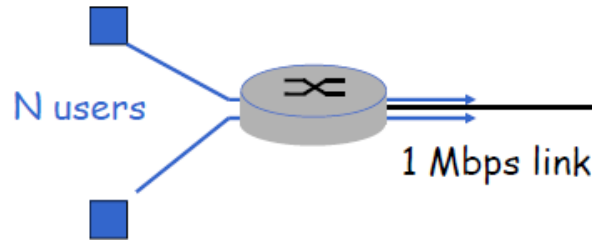
QoS: Quality of Service

More in MAC
and Transport
chapters.

May not hold
any more: VoIP

Ex.1.2 System Capacity

- **A router/switch with 1 Mbps link. There is no buffer at the router.**
- **Each user**
 - uses 100 kbps when active
 - is active 10% of the time



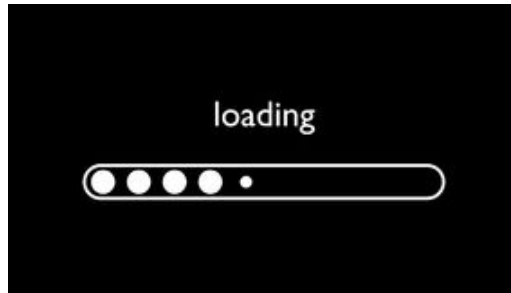
- **Find the capacity of the system**
 - Circuit switching: 10 users
 - Packet switching: ?

Probability

Packet Switching admits more users

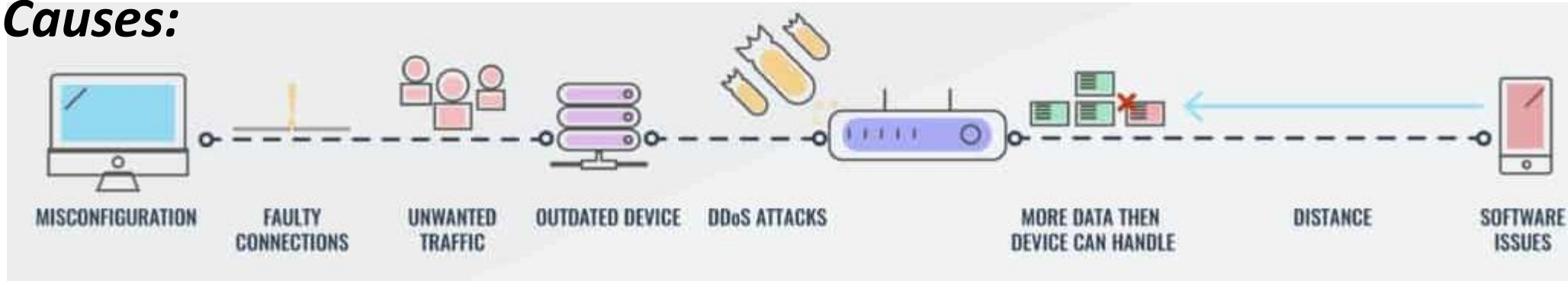
Performance of a Computer Network

- *What do you hate the most when you use the Internet?*

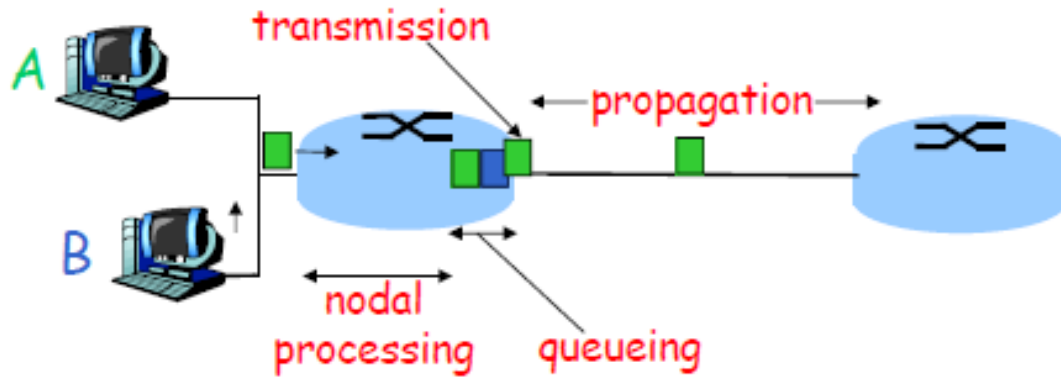


- **Delay/Latency**
- **Packet Loss**
- **Jitter**
- **Throughput**

Causes:



Delay: an Important Performance Metric



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

- **Processing delay** d_{proc}

- Checking bit error
 - Routing: deciding the output link
- Typically < ms*

- **Queuing delay** d_{queue}

- Waiting for its turn
- Value depends on congestion level

- **Transmission delay** d_{trans}

- Time to get the bit stream onto the transmission medium.
- $d_{\text{trans}} = \frac{\text{packet length}}{\text{transmission rate}} = \frac{L}{R}$
- Value depends on data size and rate

- **Propagation delay** d_{prop}

- Time for the electro-magnetic wave to reach the destination.

- $d_{\text{prop}} = \frac{\text{link length}}{\text{propagation speed}} = \frac{d}{s}$

- Typical speed s :

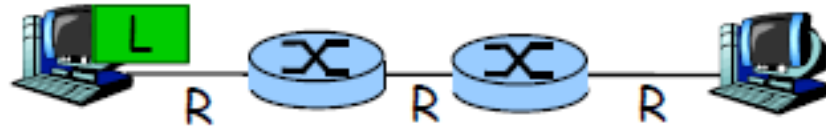
- wireless $3 \times \frac{10^8 \text{m}}{\text{s}}$
- co-ax $2 \times 10^8 \text{m/s}$

Very different!

Varies from μs to s

Ex.1.3 Delay Calculation

Consider the following lightly loaded network with 3 hops:



- Circuit setup time = x s
- Propagation delay = d s/hop
- Packet size = p bits
- Data rate at each link/hop = R bps

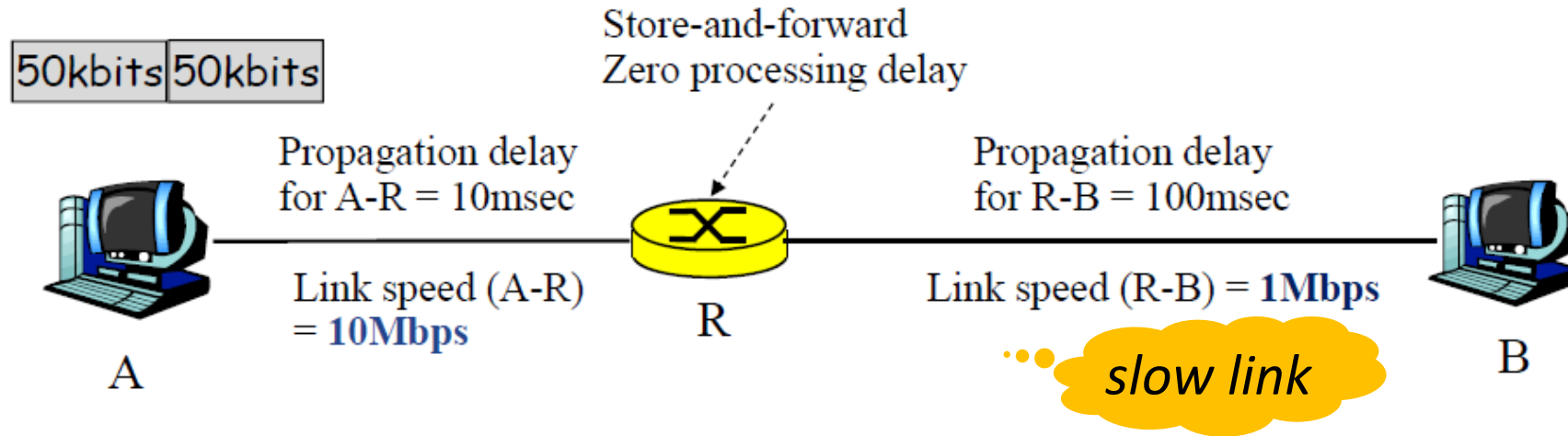
Assume there is no queuing delay or processing delay, find the total delay T for a L -bits message over the 3-hop path in

- 1) A circuit-switching network,
- 2) A packet-switching network.

Store-and-Forward

Ex.1.4 Delay Calculation-Asymmetric

Host A is to send 2 packets (back-to-back) to host B:

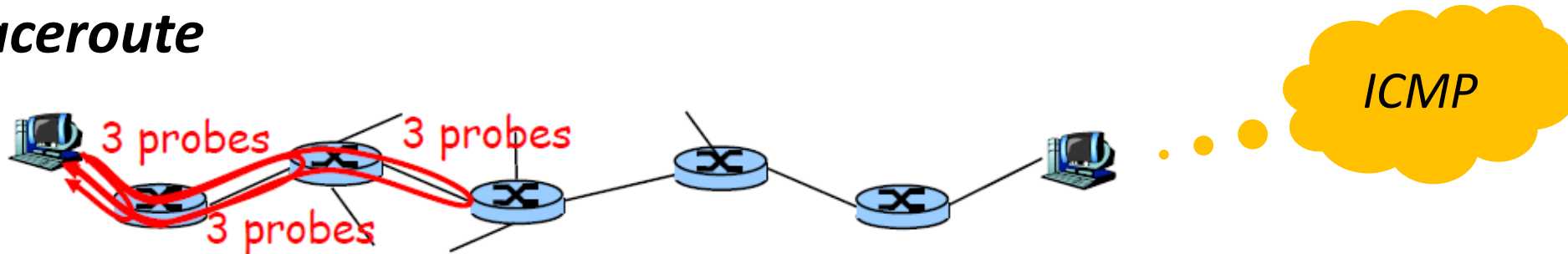


- Packet size = 50 kbits.
- Queue at the packet-switching router R is initially empty.

Find the end-to-end delay from A to B.

Delay in the real-world Internet

- **Traceroute**



>tracert www.tongji.edu.cn

通过最多 30 个跃点跟踪
到 www.tongji.edu.cn [2001:da8:b8:66:192:168:66:4] 的路由:

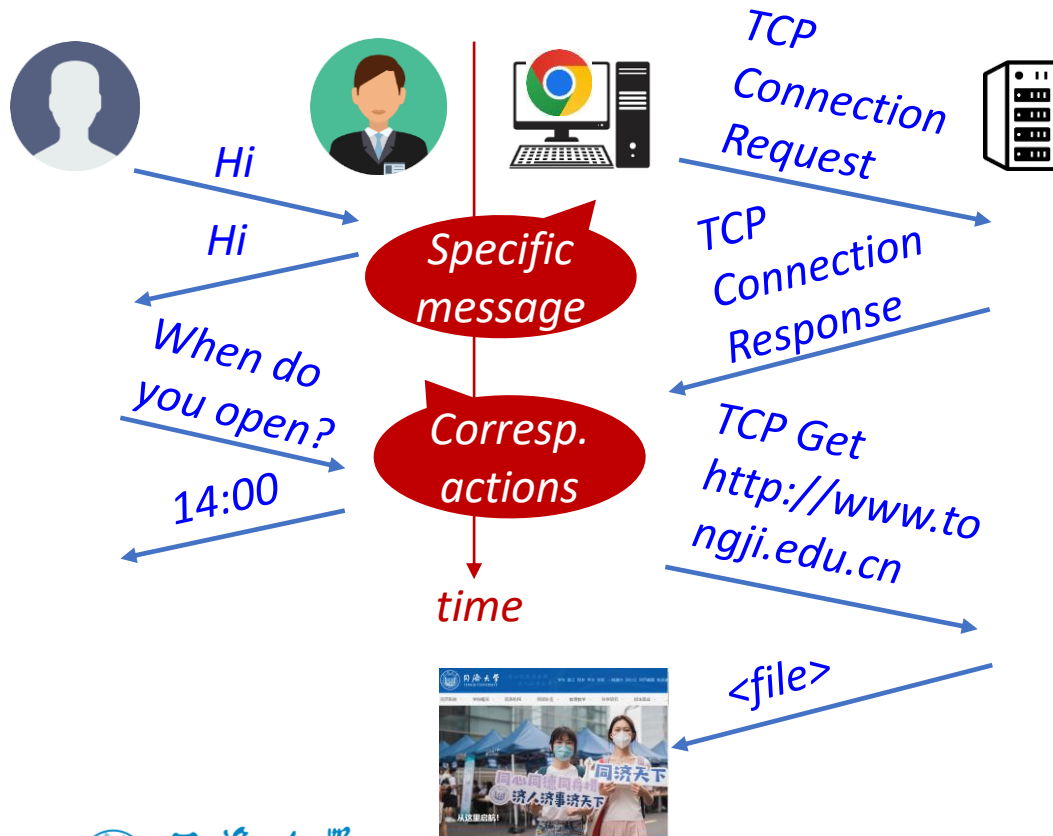
```
1      *      *      *      请求超时。
2      1 ms    1 ms    1 ms    fd00:fe::10:1
3      1 ms    1 ms    1 ms    fd01:fe::5:4
4      1 ms    1 ms    1 ms    upd-caup.tongji.edu.cn
[2001:da8:b8:66:192:168:66:4]
```

跟踪完成。

Network Software: Protocol Hierarchies

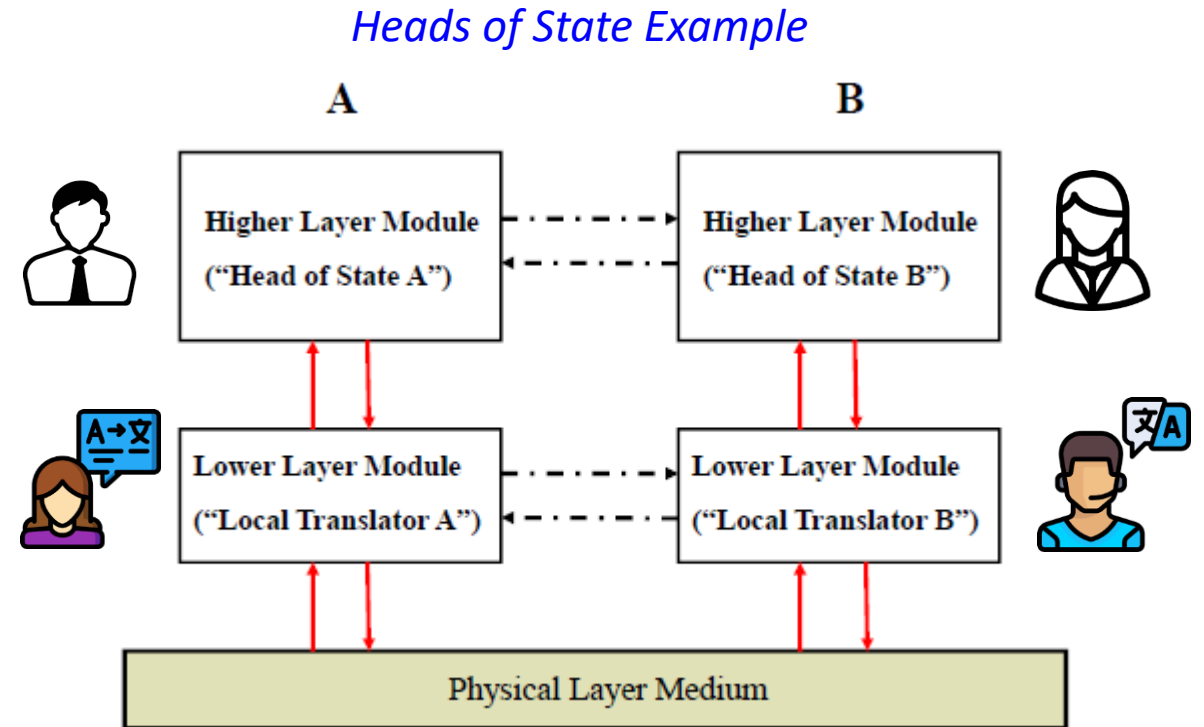
- What is a **protocol**?

A **set of rules**.



- What is a **hierarchy**?

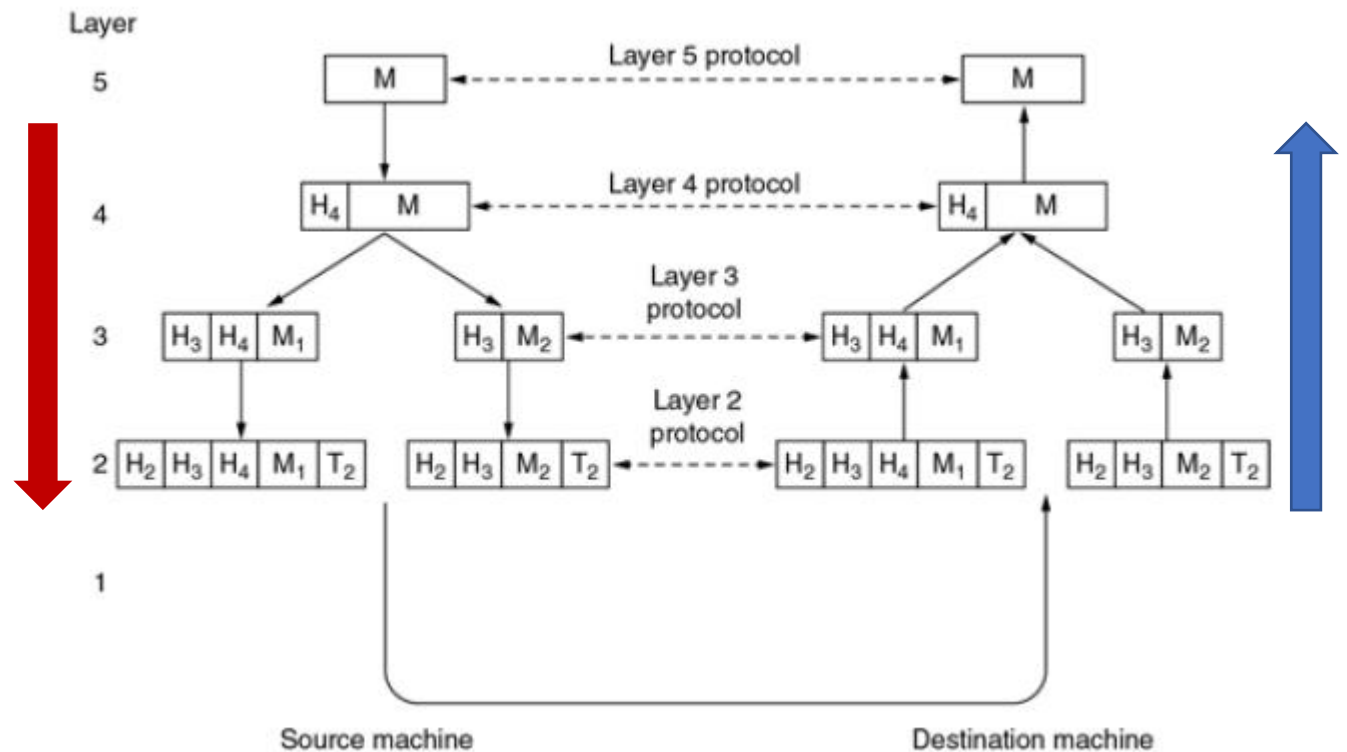
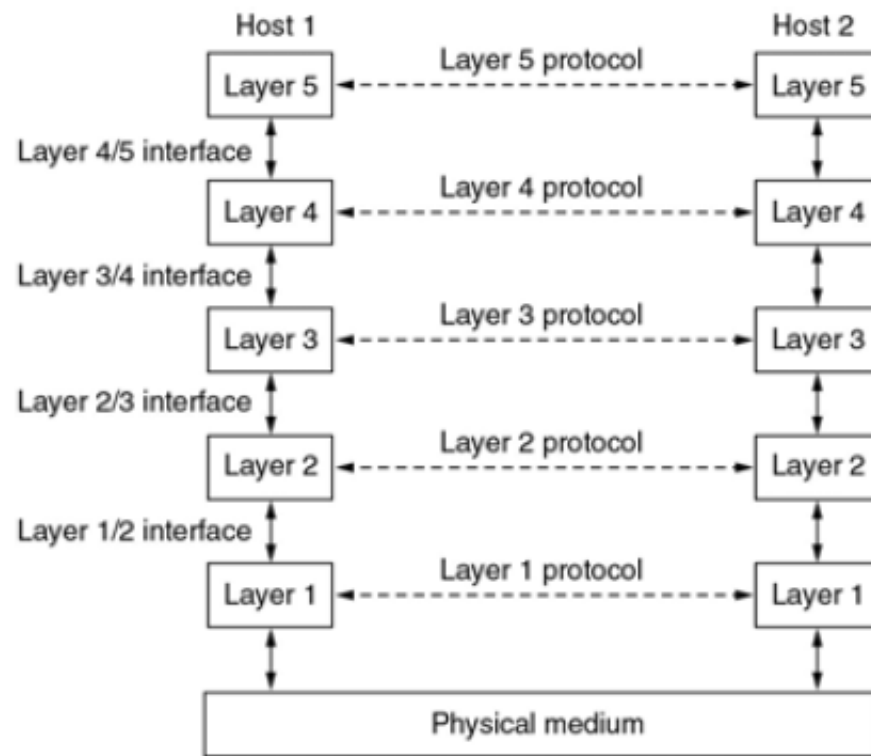
A **layered structure**.



'A new function? Add a new layer.' – anonymous programmer

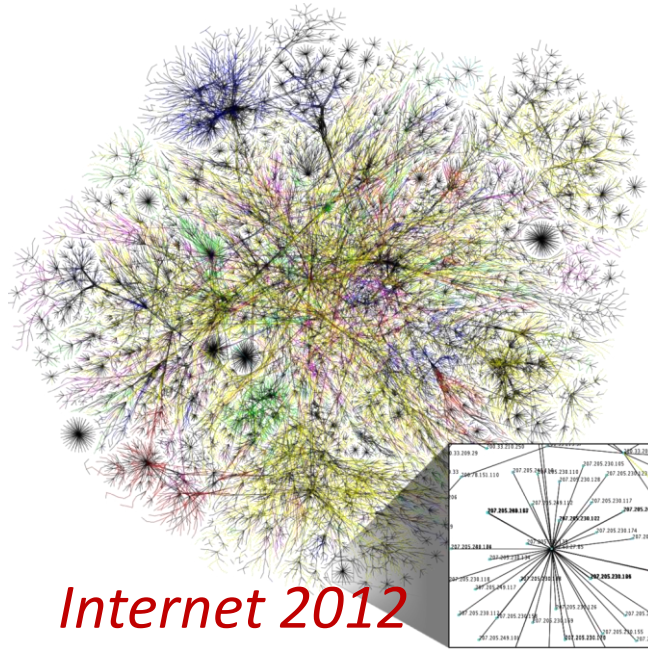
Protocol Hierarchy

- Each layer does **encapsulation**/**decapsulation**.



Why Layering?

- **Because computer network is a *complex system*:**
 - Explicit **structure** -> Easy to identify the relationship between parts.
 - **Modularization** (模块化) -> Easy to maintain/update the system.



Internet 2012

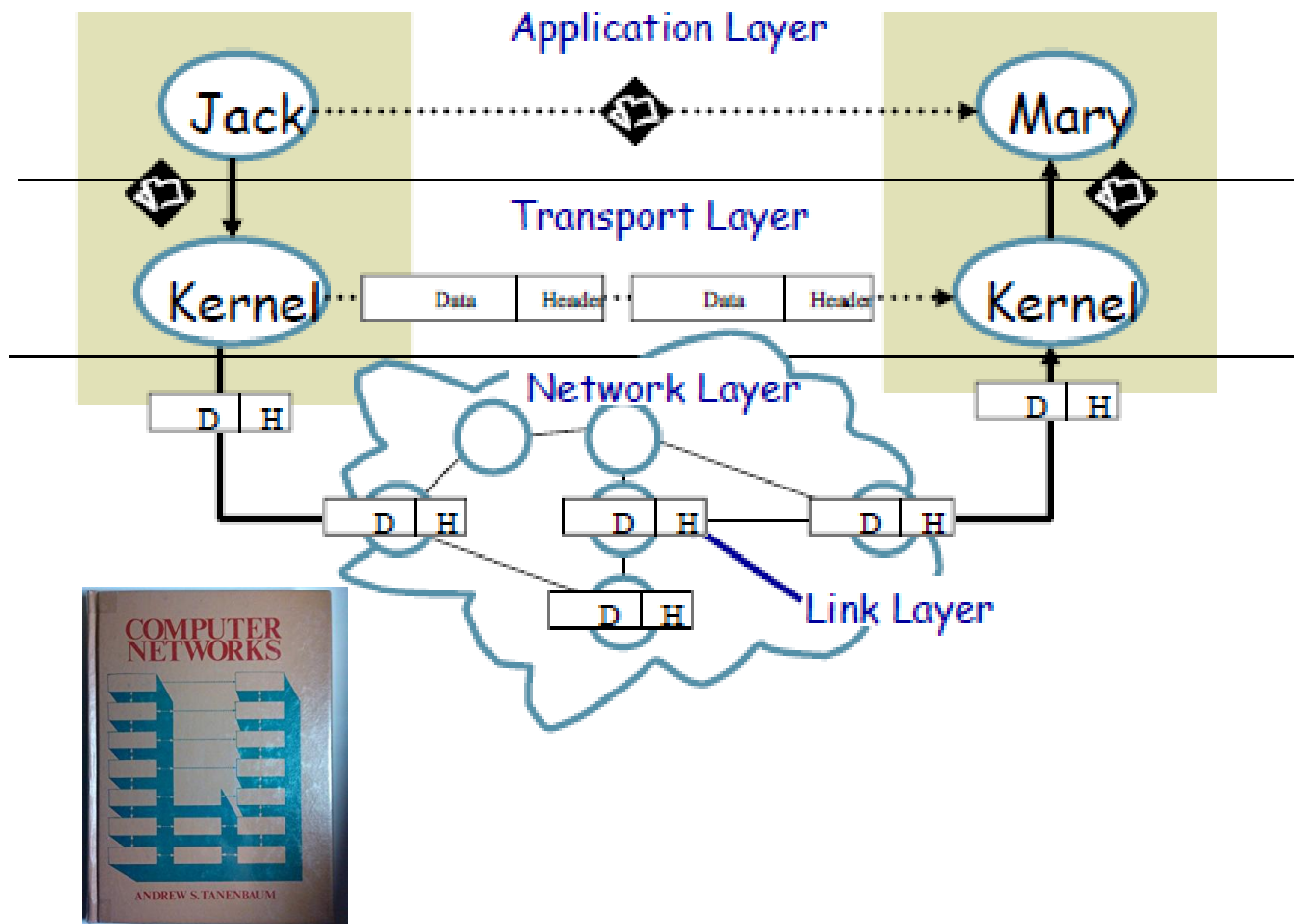
Divide and Conquer



*Horizontally: smaller
WAN – MAN – LAN*

*Vertically: finer (more layers)
Protocol Hierarchy*

A Simplified Model of the Internet



• **Packet-switched network**

- *Each packet is routed individually.*
- *No guarantee of timely delivery.*
- *No guarantee of delivery order.*
- *Even no guarantee of delivery.*
- *Data integrity may be breached.*
- *Data may be fragmented.*
- *Packets may be duplicated.*

Best-effort

What to do
with it?

Design Objectives (to be fulfilled by protocols)

- **Addressing**

- *how to identify a host/server?*



- **Routing**

- *Which way to choose for a packet to go from the source to the destination?*



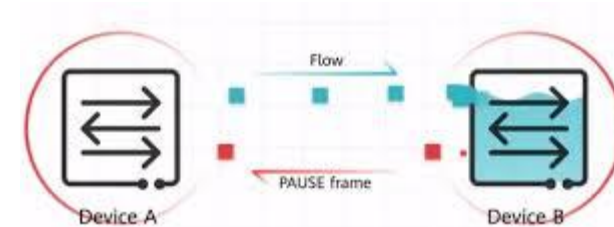
- **Error Control**

- *How to detect and correct errors?*



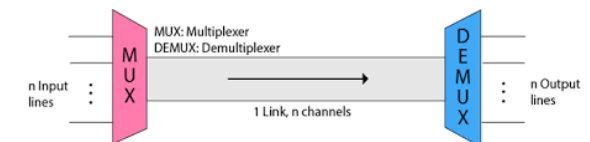
- **Flow Control**

- *How to keep a fast sender from overloading a slow receiver?*



- **Multiplexing**

- *How to share the same medium among multiple users?*



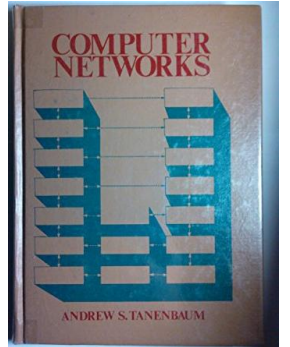
The (ISO) OSI Reference Model

Application
Presentation
Session
Transport
Network
Data Link
Physical

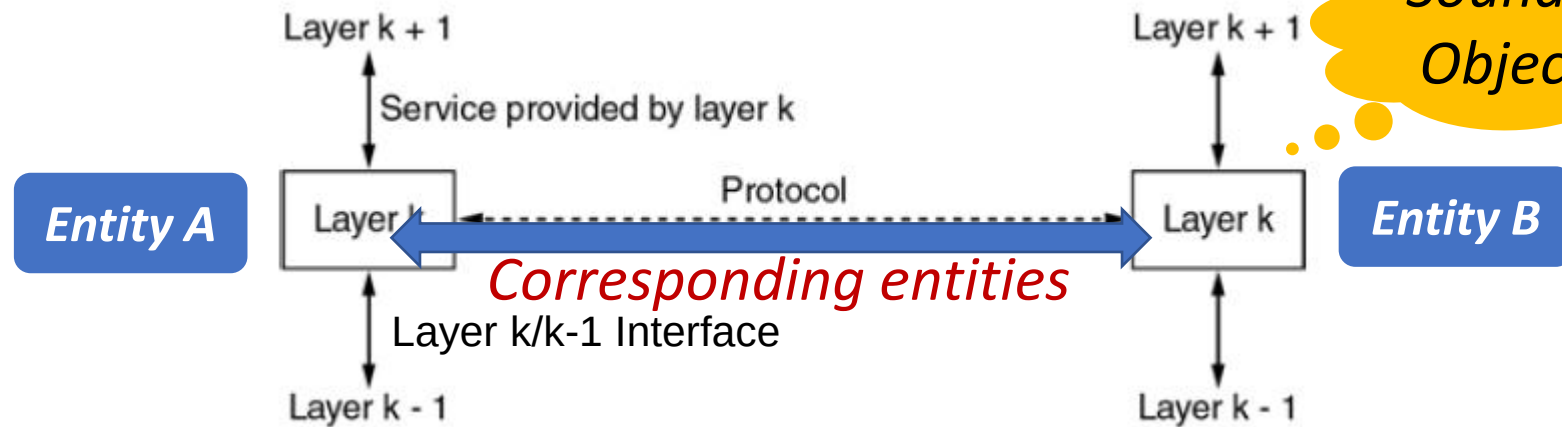
1980's, an ISO standard

- **Application**: supporting network applications
- **Presentation**: interpreting meaning of data
 - encryption/compression/syntax issue
- **Session**: session control
 - dialog control, token management, synchronization
- **Transport**: *process-to-process* reliable data transfer
 - error control, flow control, congestion control
- **Network**: routing *from source to destination*
 - subnet, addressing, routing, congestion control
- **Data Link**: transfer between *neighboring network elements*
 - (transmission) error control, framing, flow control, access control
- **Physical**: raw bits over a *communication channel*

Comprehensive,
but complicated



Protocol, Service, and Interface



implementation

- **Protocol**: a set of rules governing how communicating parties interact.

semantics

- **Service**: a set of operations that a layer provides to the layer above it.

parameters

- **Interface**: a set of allowed operations between layers/ how to access it.

The TCP/IP Reference Model

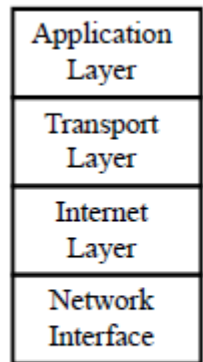
Application
Transport
internet
Link

1980's, the 'de facto' standard

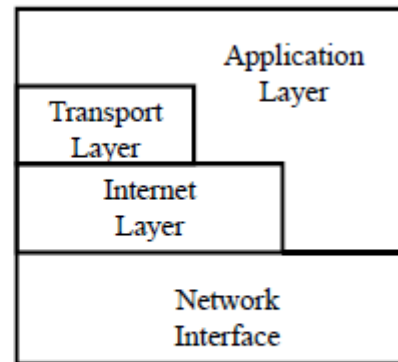
- **Application**: (OSI session + presentation + application)
 - HTTP, FTP, SMTP, TELNET, RTP, DNS
- **Transport**: end-to-end conversation
 - Transmission Control Protocol (TCP) Connection-oriented
 - User Datagram Protocol (UDP) Connectionless
- **internet**: allowing packets to go independently from source to destination (through different networks)
 - Internet Protocol (IP) + Internet Control Message Protocol (ICMP)
- **Link**: an interface between hosts and (connectionless) links
 - DSL, SONET, WiFi (IEEE 802.11), Ethernet (IEEE 802.3)

TCP/IP in the Wild (real world)

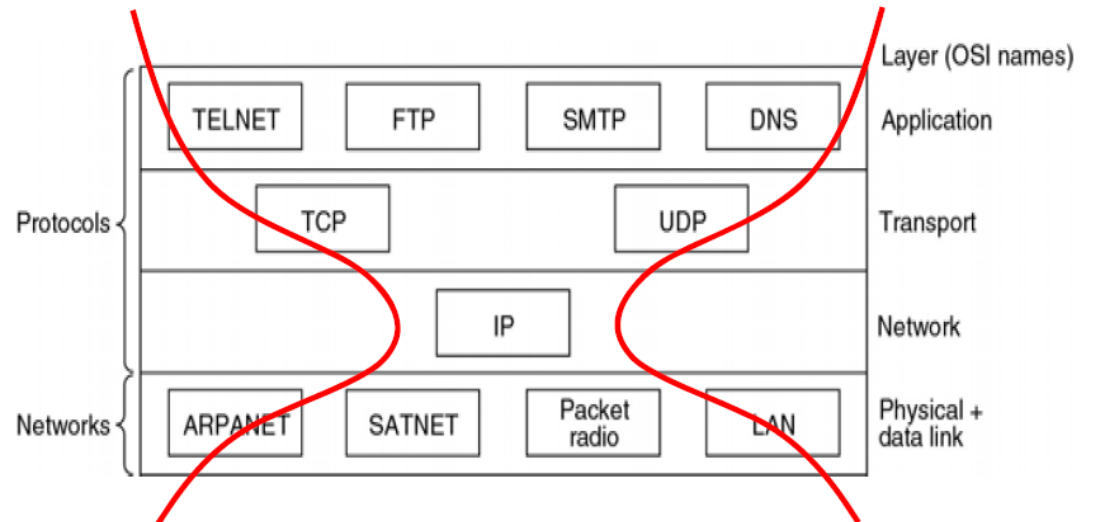
- *TCP/IP does not require strict layering, e.g. Application can bypass Transport and Internet.*
- *Everything over/below IP: it is the 'bottleneck' of the TCP/IP protocol stack.*



(a)

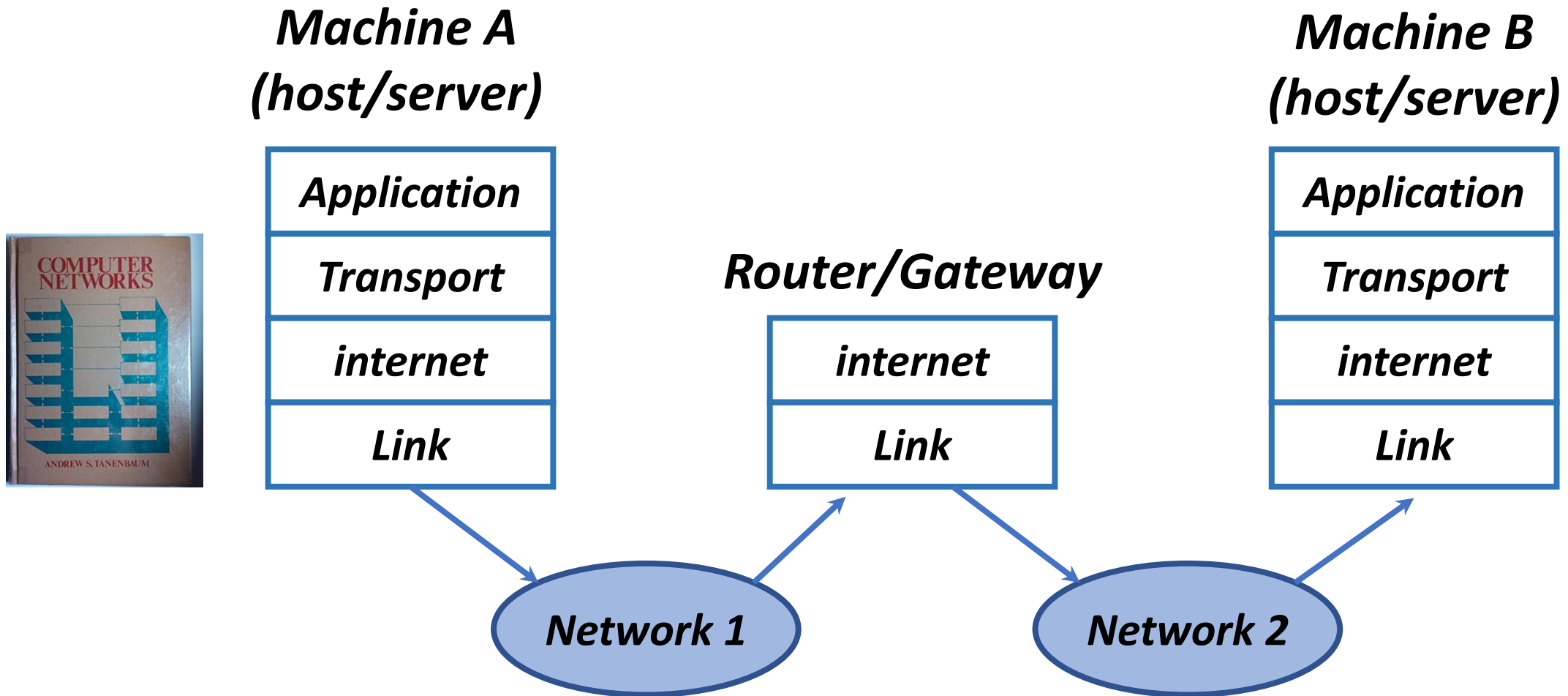


(b)



hourglass, thin-waist shaped protocol stack!!

Flow of Information across the Internet



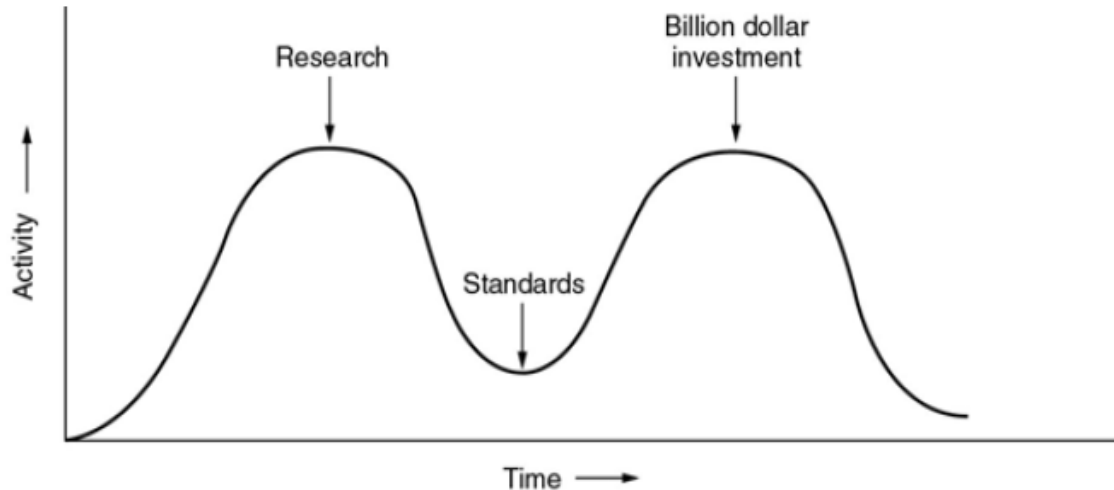
Comparison of the two reference models

- **Neither of the two reference models is perfect.**

<i>OSI Model</i>	<i>TCP/IP Model</i>
<i>Standards before protocols</i>	<i>A description of existing protocols</i>
<i>Protocol, Service, Interface</i>	<i>No clear distinction</i>
<i>Functionality not specific to layers</i>	<i>Specific functionality to layers</i>
<i>A unified architecture (which did not happen)</i>	<i>Difficult to describe other protocol stacks.</i>
<i>Connectionless/connection oriented network layer</i>	<i>Connectionless network layer</i>
<i>Connection oriented transport layer</i>	<i>Connectionless/connection oriented transport layer</i>

A Critique of the OSI Model (and Protocols)

- **Bad Timing:** investment falls behind research



- **Bad Implementations:**

huge and slow

Free

- against Berkeley UNIX
(TCP/IP implementation)

- **Bad Politics:**

- EU+ US ministries v.s.
academia

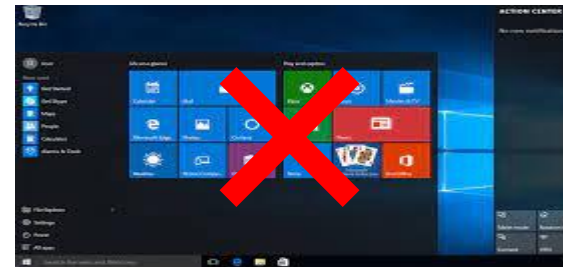
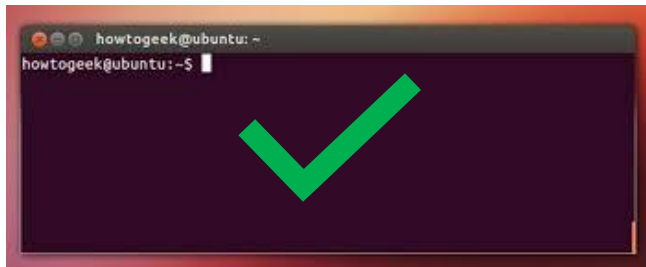
- **Bad Technology:** political rather than technical

- Empty presentation and session layer
- Difficult to understand or implement



A Critique of the TCP/IP Model

- Does not distinguish protocol/service/interface
- *Not a general model*
- Link layer is actually a 'host-to-network' interface, not a layer
- Does not distinguish physical and data link layers
- TCP/IP are well developed and implemented, but other *protocols* are not. (What's worse) They are *deeply entrenched, and hard to replace*.
 - E.g. TELNET, designed for 10-char-per-second mechanical terminals, still in use



A hybrid reference model in this course

Application
Transport
Network
Link
Physical

- **Application**: supporting network applications
 - HTTP, FTP, SMTP, DNS, TELNETmessage
- **Transport**: *process-to-process* data transfer
 - TCP, UDPreliable,
orderedpacket
- **Network**: routing *from source to destination*
 - IP, and other routing protocolsbest-effortdatagram
- **Link**: data transfer between *neighboring network elements*
 - Ethernet (IEEE 802.3 MAC), WiFi (IEEE 802.11 MAC)frame
- **Physical**: raw bits over *communication channel*
 - Ethernet (IEEE 802.3 PHY), WiFi (IEEE 802.11 PHY)bits

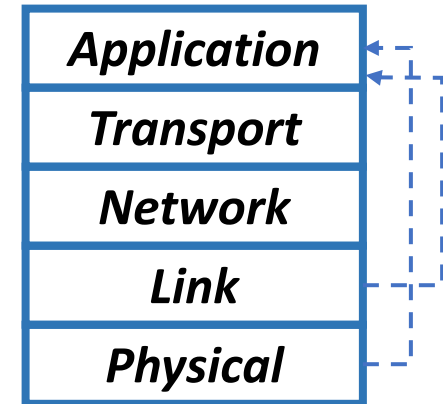
Cross-layer Design?

- **WHY:** *wireless/ad-hoc networking*

1980s

After 2000

- The layered principle was invented when wireless network was not popular.
- Cross-layer design almost always improves the performance because
 - *More information* is available to the decision maker.
 - Eliminate some procedure.
- In wireless network, performance matters (and downgrades fast).

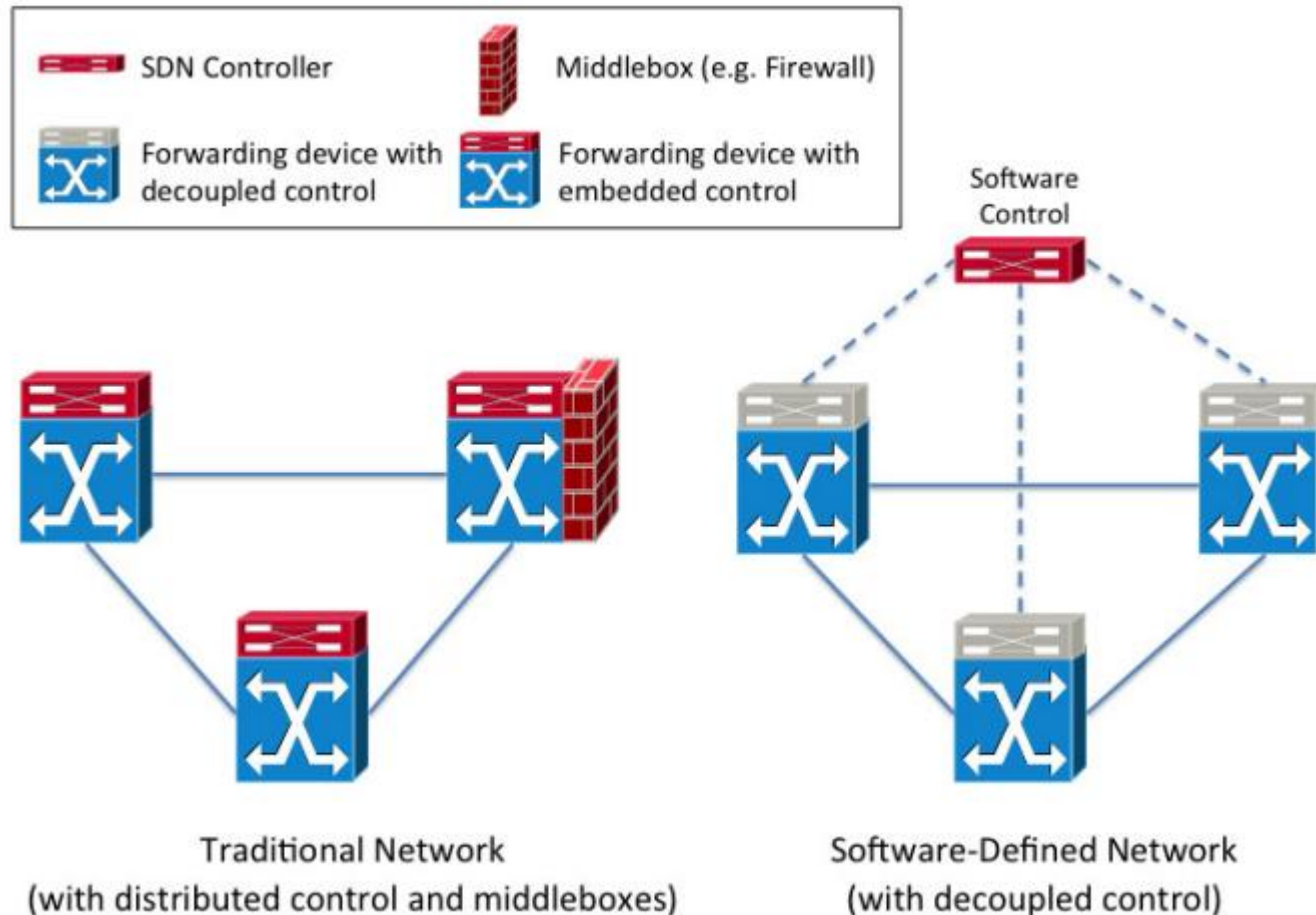


- **What are the possible problems?**

Very popular
around 2010

Example: Software Defined Network (SDN)

- **Network Function Virtualization**



History of Internet (1/4)

- *1961-1972: Early packet-switching principles*
 - *1961: Kleinrock – queueing theory shows effectiveness of packet-switching*
 - *1964: Baran – **packets witching** in military networks*
 - *1967: ARPAnet conceived by Advanced Research Projects Agency*
 - *1969: first ARPAnet node operational*
 - *1972: ARPAnet public demonstration, first email program, first host-host protocol (Network Control Protocol, NCP)*

History of Internet (2/4)

- *1972-1980: Internetworking, new and proprietary networks*
 - *1970: ALOHAnet satellite network in Hawaii*
 - *1974: Cerf and Kahn - **architecture for interconnecting networks***
 - *1976: **Ethernet** at Xerox PARC (also GUI, laser printing)* *'Minimalism, autonomy, best-effort service model, stateless routers, decentralized control'*
 - *Late 70's: proprietary architectures: DECnet, SNA, XNA*
 - *Late 70's: switching fixed length packets (ATM precursor)*
 - *1979: ARPAnet has 200 nodes*

Defines today's
Internet
architecture

History of Internet (3/4)

- *1980-1990: new protocols, a proliferation of networks*
 - *1983: deployment of **TCP/IP***
 - *1982: SMTP e-mail protocol defined*
 - *1983: DNS defined for name-to-IP-address translation*
 - *1985: ftp protocol defined*
 - *1988: **TCP congestion control***

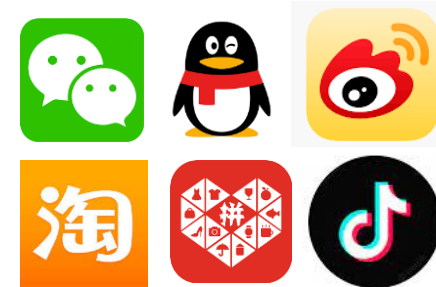
History of Internet (4/4)

- *1990, 2000's: commercialization, the Web, new apps*
 - *Early 1990's: ARPAnet decommissioned -> **Internet***
 - *Early 1990s: Web*
 - *hypertext [Bush 1945, Nelson 1960's]*
 - *HTML, HTTP*
 - *Late 1990's: Commercialization of the Web*
 - *Late 1990's to 2000's: New applications*
 - *Instant Messaging, P2P file sharing*
 - *Network security*
 - *50 Mil. Host, 100 Mil. Users*
 - *Internet Backbone (core) links running at Gbps*
- *Internet access becomes a utility, like electricity and water.*



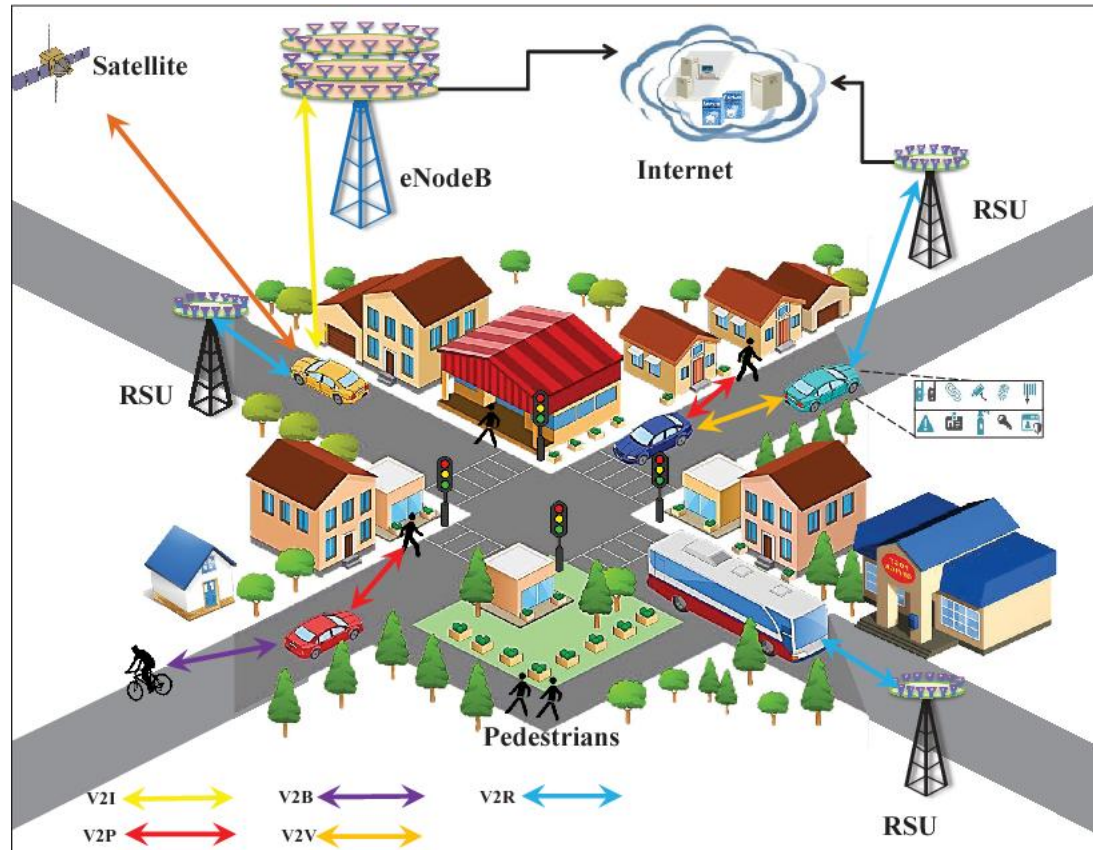
Mobile Internet (after 2005)

- **Smartphones**
- **Aggressive deployment of broadband access**
- **High-speed wireless access -> mobile applications**
 - *Social networks: Facebook, Wechat, Weibo*
 - *E-commerce: Taobao, Amazon, PDD*
 - *Payment and Banking: Alipay, Wechatpay*
 - *Video and Streaming: Youtube, Netflix, Tik Tok*
 - *Gaming: Tencent games, miHoYo*
 - *Education: coursera*



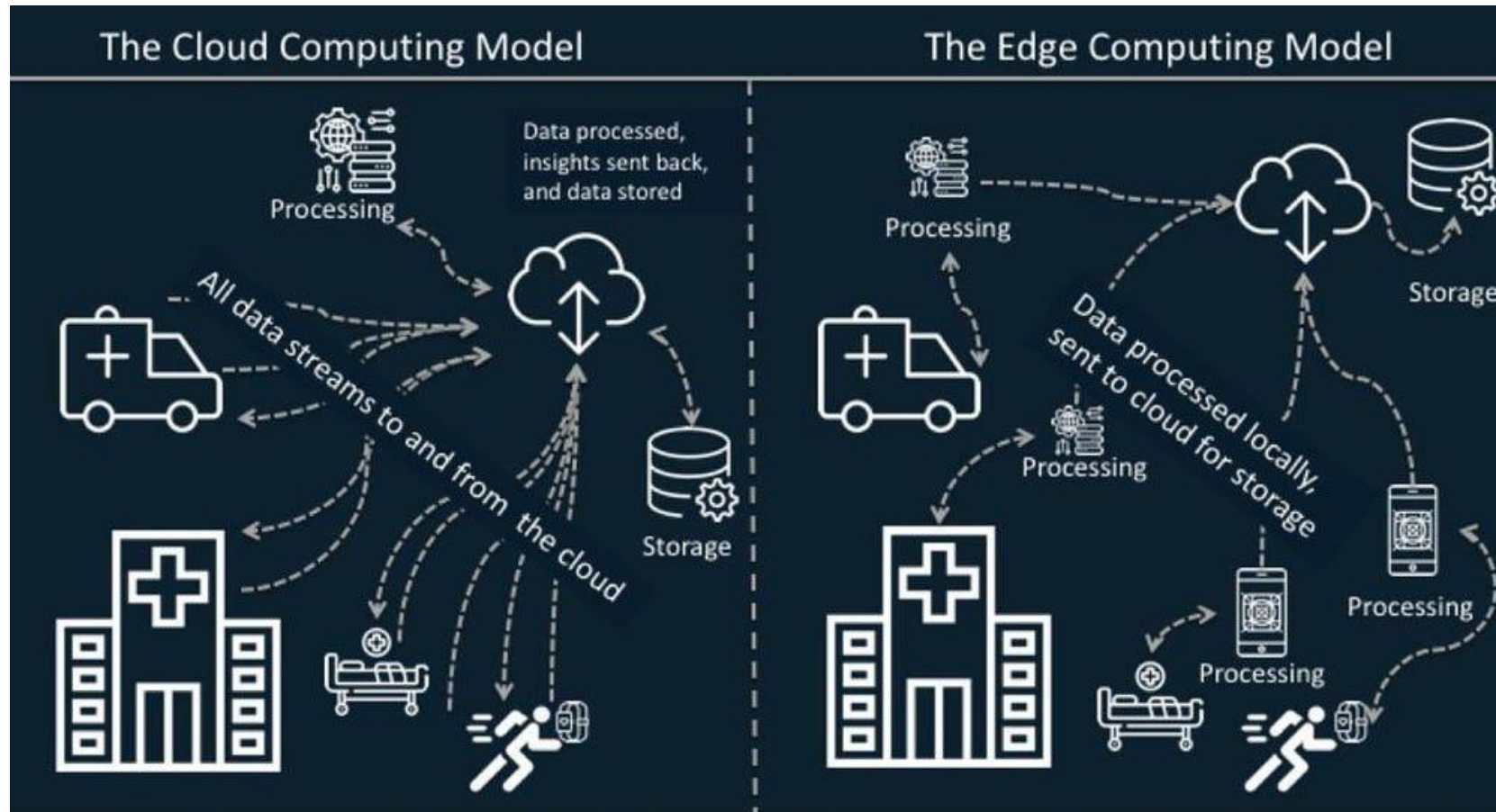
Future/Ongoing Directions (1/2)

- ***Vehicle Network (Internet of Vehicles)***



Future/Ongoing Directions (2/2)

- **Edge Computing Network**



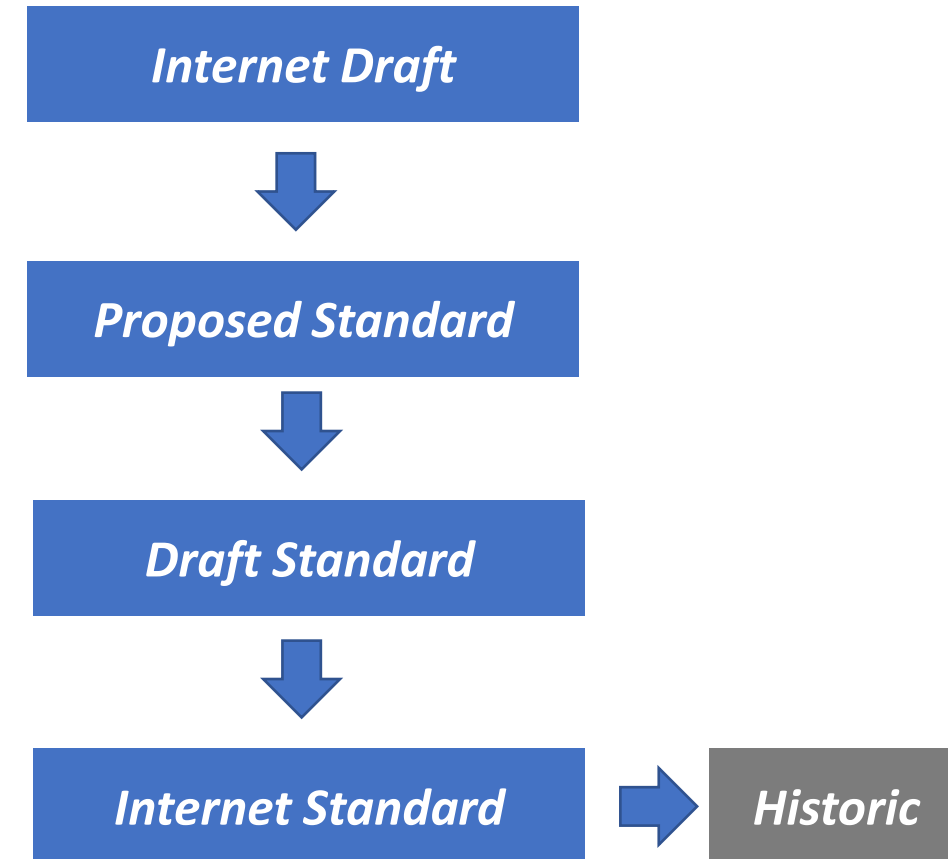
Standardization

Network Standards Organizations

ANSI	American National Standards Institute
IEEE	Institute of Electrical and Electronics Engineers
ITU	International Telecommunications Union
ISO	International Organization for Standardization
ISOC	Internet Society
IETF	Internet Society and the Internet Engineering Task Force
EIA/TIA	Electronic Industries Alliance and the Telecommunications Industry Association

- **3GPP – cellular networks**

- ***Request for Comment (RFC)***



Summary

- ***Classification of computer networks by scale/range***
- ***Frequently used technologies***
 - *In access network: ADSL, FTTH, WiFi, 5G...*
 - *In core network: IP, frame relay, ATM...*
- ***Packet switching v.s. Circuit switching***
 - *Store-and-forward*
- ***Performance of computer networks, QoS***
- ***Reference Models: OSI v.s. TCP/IP***
 - *What does each layer do?*